

DEPI Junior Data Analyst Track

Graduation Project

Team Members: 6

Date: October 2026

UK Train Rides Data Analysis and Forecasting

Project Proposal

1. Project Overview

This project focuses on analyzing UK train ride data using Power BI in order to understand travel patterns, ticket demand, revenue performance, and other important trends within the railway dataset.

The project will include data cleaning and preprocessing, exploratory data analysis, business-oriented analysis questions, interactive data visualization, and forecasting.

The final solution will provide an interactive Power BI dashboard that presents the main findings and supports data-driven decision making.

2. Problem Statement

Railway travel data contain valuable information about passenger journeys, ticket purchases, revenue, travel routes, and service performance.

However, raw data alone does not provide clear insights for decision makers.

The main problem addressed by this project is how to transform the available railway dataset into meaningful information that can help understand travel demand, revenue trends, ticket purchasing behavior, and future demand.

The project will also use historical data to estimate future train ride demand and revenue.

3. Project Objectives

The main objectives of the project are:

1. Clean and preprocess the UK train rides dataset using Power BI.
2. Understand the structure, quality, and main characteristics of the data.
3. Identify meaningful analysis questions that can support decision makers.

4. Analyze the number of train journeys and travel patterns.
 5. Analyze revenue performance across different periods and categories.
 6. Analyze passenger ticket preferences and ticket classes.
 7. Identify important patterns and trends within the railway data.
 8. Predict the number of rides for the next month.
 9. Estimate the forecasted revenue for each day of the next month.
 10. Identify the expected demand across different ticket classes.
 11. Build an interactive Power BI dashboard that clearly presents the analysis results.
 12. Prepare final project documentation, report, and presentation.
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4. Project Scope

In Scope

The project will include:

- Data cleaning and preprocessing
- Data transformation using Power Query
- Data modeling in Power BI
- Exploratory data analysis
- Travel and journey analysis
- Revenue analysis
- Ticket class analysis
- Identification of trends and patterns
- Forecasting of future train rides
- Forecasting of daily revenue
- Analysis of ticket class demand
- Interactive Power BI dashboard
- Project documentation
- Final report and presentation

Out of Scope

The project will not include:

Development of a website or mobile application

Backend development

API development

Deployment of a software system

Advanced machine learning development outside the requirements of the Junior Data

Analyst project

5. Tools

The primary tool used in this project will be:

Microsoft Power BI

Power BI will be used for:

Data cleaning

Data transformation

Data modeling

Data analysis

Forecasting

Data visualization

Dashboard development

6. Expected Deliverables

The expected project outputs are:

A cleaned and prepared railway dataset

A documented set of business and analysis questions

A Power BI data model

Analysis results and insights

Train ride forecast for the next month

Forecasted daily revenue for the next month

Ticket class demand analysis

Interactive Power BI dashboard

Final project report

Final presentation

7. Project Team

Team Size: 6 Members

Member 1: John George

Member 2: Sandy Saher

Member 3: Farah Magdy

Member 4: Ammar Ashraf

Member 5: Shahd El Bhrawey

Member 6: Rahma Mohamed

8. Project Repository

GitHub Repository:

<https://github.com/johngeorge2006/UK-Train-Rides-Data-Analysis-Forecasting>